The Personal Media Intelligence Hub

A Single-User, Cross-Domain Taste-Transfer Study with Honest, Frozen-Fold Evaluation

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Single-author technical report — for the author's own records

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Abstract

The Personal Media Intelligence Hub (PMIH) asks a deliberately hard version of the recommendation problem: can a *single* person's taste, learned from a few thousand ratings across five media domains (movies, TV, games, books, music), *transfer* from one domain to another? The project ingests five domains through seven content APIs plus Wikidata, projects them into a shared MiniLM→PCA semantic space, and evaluates every claim on one frozen 50-fold registry with paired Wilcoxon tests. Three findings anchor the report. (1) *Transfer into Games is positive and survives adversarial controls*: the `movie+book→game` configuration lifts augmented skill by +0.121 at full target data (paired $p = 0.0015$; significant at 3/4 data fractions) and remains content-driven after a domain-identity leak fix, a prior-vs-transfer control (Spearman +0.368), and a text-length normalization control [`transfer_grid_summary.json`; `PIPELINE_RUN_LOG.md`]. (2) *Feature-level geometry does not predict transfer*: centroid distance vs. zero-shot skill has Spearman $\rho = -0.18$ over 12 domain pairs [`transfer_grid_summary.json`]. (3) *The custom edge-penalty XGBoost loss, freshly ablated here, is a near-no-op at its tuned operating point*: on the movies domain it changes overall MAE by −0.0020 (paired $p = 0.21$) and edge-MAE by −0.0097 (paired $p = 0.28$), both non-significant [`edge_loss_ablation_results.json`]; an α-strength sweep shows the mechanism works as designed (edge-MAE falls, spread widens) but trades against overall accuracy, which is why Optuna tuned α small. The report also treats self-correction as a contribution: a benchmark-inversion bug (5× N inflation), a leak-faked "significant" bridge result, and stale-artifact drift were each caught and fixed.

Contents

1 Introduction

Mainstream recommender systems exploit *many* users to cover for sparse per-user signal. This project removes that crutch: there is exactly one user, a few thousand ratings, and four of five domains with $N < 200$. The motivating question is whether *taste* — the latent function mapping content to a 0.5–5.0 star rating — is shared enough across media that data from a data-rich domain (movies, $N = 980$) can help a data-poor one (games, $N = 62$).

Why single-user / small-N is the hard version. With one user there is no collaborative signal, no implicit co-occurrence across users, and no way to average out idiosyncrasy. Every metric is therefore computed against the only honest baseline available — predicting the user's mean rating — via a *skill score* (Section 10). Small N also makes the evaluation protocol, not the model, the dominant risk: a single careless split inflates or inverts results. The project's response is a single frozen-fold registry reused by every experiment.

Contributions.

1. A reproducible, frozen-fold, single-source-of-truth evaluation harness for single-user multi-domain rating prediction (Section 10).
2. A transfer study with pre-registered decision rules and three adversarial controls, yielding one **[CONFIRMED]** positive finding (`movie+book`→`game`) and an explicit feature-geometry null (Section 7).
3. The first head-to-head ablation of the project's custom edge-penalty loss, run for this report, with an honest non-significant verdict and a mechanism sweep (Section 5).
4. A catalogue of methodological failures caught and corrected, presented as a contribution rather than hidden (Section 10).

2 Background and Related Work

The cold-start problem — predicting for items or users with little history — is classical in recommendation [1]. With a single user, *every* item is effectively a cold-start in the collaborative sense, so the system leans entirely on content features. Implicit-feedback modelling [2] is relevant to the music domain, where explicit stars are absent and engagement signals (library membership, play counts, playlist inclusion) stand in for ratings. Treating "in library" as positive and everything else as unlabeled is positive-unlabeled (PU) learning [3].

Cross-domain knowledge sharing is transfer / multi-task learning [4, 5]. The shared semantic space is built from Sentence-BERT embeddings [6] reduced by PCA; domain shift in that space is attacked with CORAL covariance alignment [7]. The base learners are gradient-boosted trees [8]. Uncertainty is quantified with split conformal prediction [9], which guarantees marginal coverage without distributional assumptions. All significance tests use the paired Wilcoxon signed-rank test [10] on per-item absolute errors, and the latent space is visualised with UMAP [11].

3 Data and Feature Engineering

Domains and sources. Five domains are ingested through seven primary content APIs — TMDB (movies, TV), RAWG (games),[1] Google Books (books), Spotify, MusicBrainz, Genius and ReccoBeats (music) — plus Wikidata for cross-domain entity linking (Section 8) and a sixth, YouTube, ingested but not modelled (Section 12). All API responses are cached on disk (`data/cache/`) so re-runs are deterministic and cheap.

Per-domain corpus. Table 1 gives N, rating mean and variance per domain, computed directly from the unified training frame [`data/processed/unified/training_features.csv`]. A rating-date span column is deliberately omitted: `rating_date` is a placeholder in this frame — it is NaN-filled to `2000-01-01` for every row [`unified_feature_engineering.py`:272], so no real date range exists (this has consequences for the temporal-decay claim; see Section 6). Figure 1 shows the rating histograms: ratings cluster tightly around 3–4 stars in every domain (movie SD 0.96, music SD 0.57), which is the low-variance problem that motivates the skill score and the edge-penalty loss.

Table 1: Per-domain corpus summary [`training_features.csv`].

Domain	N	rating mean	rating var
Movies	980	3.24	0.93
TV	159	3.45	1.17
Games	62	3.39	2.44
Books	63	3.48	0.62
Music	3688	4.18	0.33

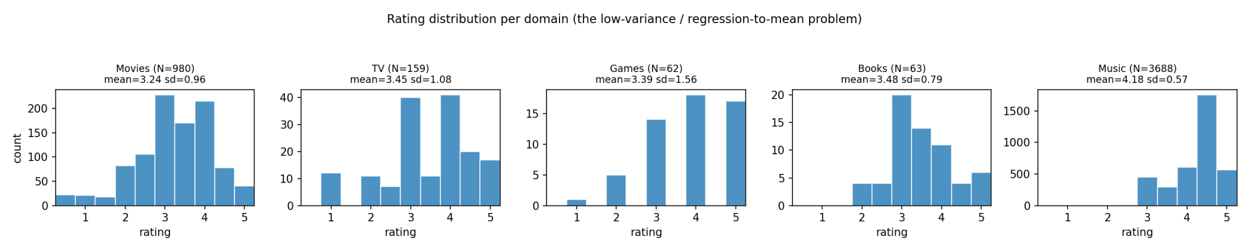

Figure 1: Rating distribution per domain. Low variance and a 3–4 star mode are the central difficulty; predicting the mean is a strong baseline.

Shared semantic space. Each item's title, lead creator and overview/description are embedded with `all-MiniLM-L6-v2` (384-dim) [6] and reduced to **10 PCA "vibe" components** in the unified space [`config.py`:133; `unified_feature_engineering.py`]. MiniLM was chosen for its small size and strong sentence-level semantics on short text; PCA (rather than a learned projection) keeps the transform fit-once, inspectable, and stable at small N, where a trained encoder would overfit. Per-domain standalone models use larger PCA budgets (movies 25, music 32, games/books/TV 15) because they are not constrained to a common basis.

Genre unification and critic scores. Genres are unified across domains with a `MultiLabelBinarizer` (e.g. `gen_Action` fires for movies, TV and games; 16 genres span ≥ 2 domains [`PIPELINE_RUN_LOG.md` Check 5]) and capped at the 40 most frequent in the transfer grid [`transfer_study.py`]. Heterogeneous critic scores (IMDb, Metacritic, Rotten Tomatoes, TMDB) are normalised to a single `critic_avg_5` on a 0–5 scale. Figure 2 shows that coverage of the shared features is uneven — the shared space is genuinely thin, which bounds how much cross-domain pooling can help.

[1] Code-vs-doc discrepancy: prose docs occasionally reference IGDB for games, but the implemented ingestion uses RAWG (`src/games/ingestion.py`). Trusting the code per the project's measurement-honesty rule.

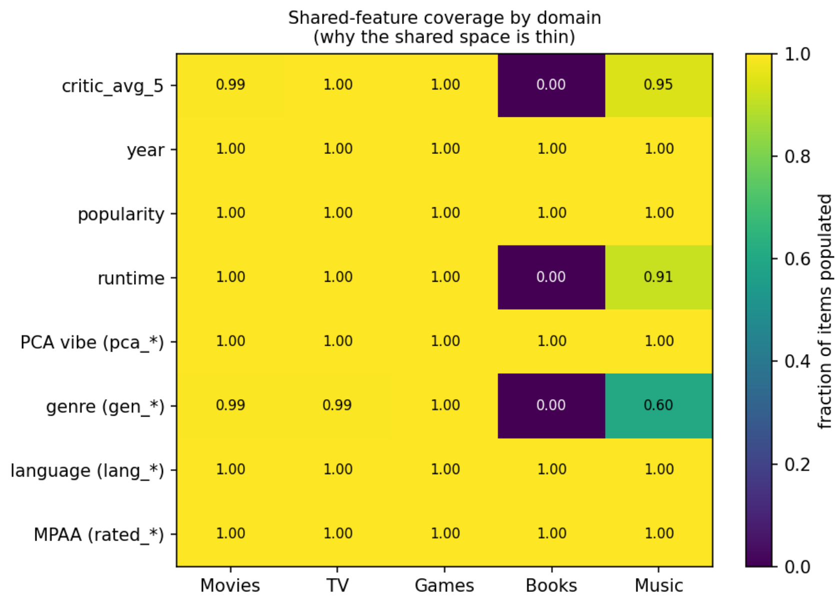

Figure 2: Shared-feature coverage by domain (fraction of items with a populated value). The PCA vibe space and genre one-hots are dense; numeric metadata is domain-specific.

CORAL alignment. Domains occupy different regions of the embedding space (raw separability ratio 1.25); per-fold centroid/CORAL alignment [7] to a movie reference collapses this to ≈ 0 [`PIPELINE_RUN_LOG.md` Check 3; `unified_utils.py`:15–75], so that cross-domain proximity reflects content rather than domain identity (Figure 3, left). Alignment is always fit on training folds only. The same diagnostics flag a text-shape disparity (Figure 3, right): median description length ranges from 214 chars (movies) to 1074 (games), motivating the text-normalization control in Section 7.

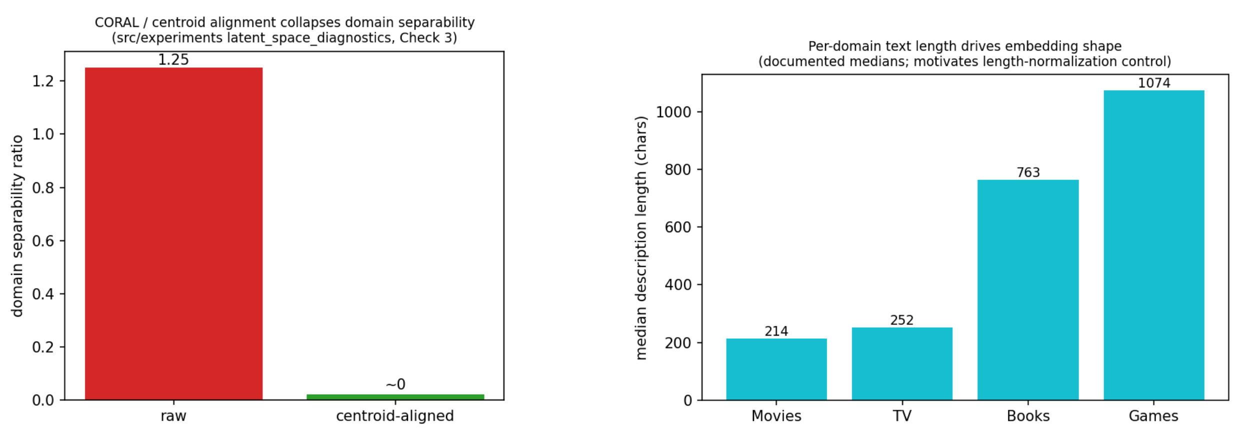

Figure 3: Left: centroid/CORAL alignment collapses domain separability ($1.25 \rightarrow\approx 0$). Right: per-domain median description length (documented medians from the latent-space diagnostics).

Latent space, visualised. Figure 4 projects the shared space with UMAP [11]: coloured by domain it shows clear stratification; coloured by rating it shows only a weak taste gradient — consistent with the low per-domain R^2 and the aleatoric-ceiling argument.

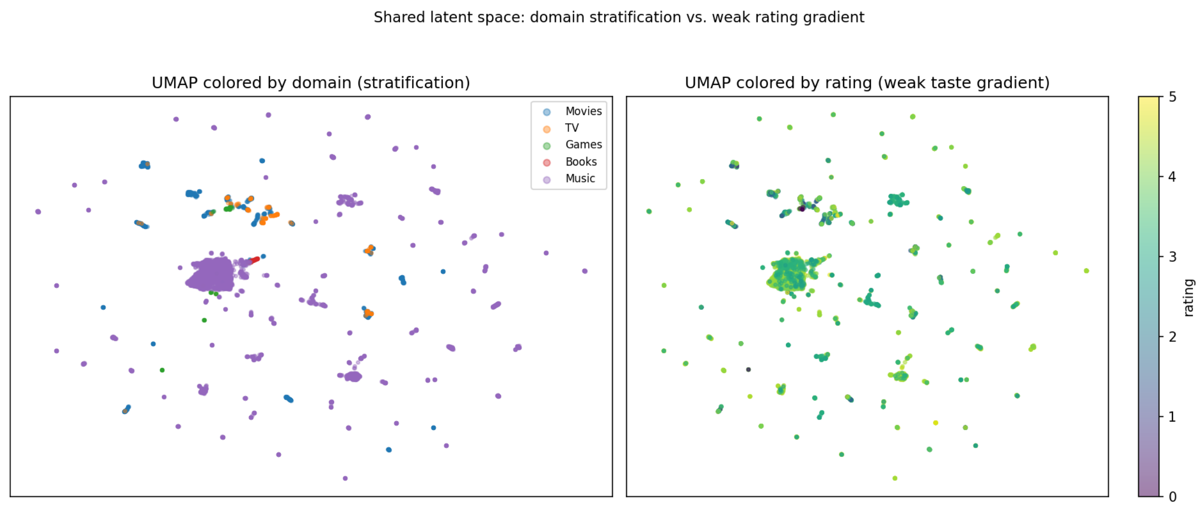

Figure 4: UMAP of the shared space: domain stratification (left) vs. a weak rating gradient (right) [`training_features.csv`].

4 Per-Domain Models

Table 2 is the honest final benchmark: each domain's own deployed model on the 50 frozen folds, with N counted as *unique items* (not OOF rows). Skill is $1 - \mathrm{MAE}_{\mathrm{model}}/\mathrm{MAE}_{\mathrm{mean}}$, computed on raw OOF predictions for the production movie/TV stacks and the local game/book SVRs.

Table 2: Per-domain benchmark on 50 frozen folds [`latest_metrics.json`; skill derived from `production_oof.csv` / `standalone_oof.csv`]. Predictions clipped to $[0.5, 5]$ and rounded to the nearest half-star before scoring.

Domain	N	Model	MAE	R^2	Skill	$\pm 0.5\star$ Acc
Movies	980	Prod Stacking (edge+Cat+SVR+Ord)	0.478	0.558	0.362	79.1%
TV	159	Prod Simplex-Stack (edge+Cat+SVR)	0.827	0.052	0.075	53.5%
Games	55	Local SVR	0.636	0.364	0.207	61.8%
Books	63	Local SVR	0.548	0.182	0.126	76.2%

Movies ($N = 980$). The production model is a leakage-safe stack of an edge-penalty XGBoost regressor, CatBoost, an SVR, and an ordinal expected-value head, with a `RidgeCV` meta-learner [`src/reporting/production_benchmarks.py`]. It reaches MAE 0.478, $R^2 = 0.558$, 79.1% within $\pm 0.5\star$ [`latest_metrics.json`] — the strongest domain by far, because it has the most data and the richest metadata.

TV ($N = 159$). A simplex-weighted stack (XGB+CatBoost+SVR, non-negative weights summing to one) reaches MAE 0.827, $R^2 = 0.052$ [`latest_metrics.json`]. The near-zero R^2 is honest: TV ratings carry little signal the features can explain, and the heavy ensemble does not rescue a low-signal domain (the simpler proxy scored $R^2 = 0.063$ — [`PIPELINE_RUN_LOG.md`]).

Games and Books ($N = 62/63$). Both use a local `SVR(rbf)` with frozen hyperparameters (C=1.0, ϵ=0.1), chosen over boosted trees because at $N < 70$ a high-variance learner overfits. Games MAE 0.636 ($R^2 = 0.364$), Books MAE 0.548 ($R^2 = 0.182$) [`latest_metrics.json`]. The cross-domain *distillation prior* was tested as an added feature here and dropped (Section 6). [2]

[2]Games appears as $N = 62$ in Table 1 (corpus) but $N = 55$ in Table 2 (modelled): the corpus holds 62 games, of which 55 carry usable ratings after excluding 7 "Incomplete"-status items with no target [`standalone_benchmarks.py`]. Both counts

Music. Music has no explicit stars, so a continuous 1.0–5.0 implicit rating is synthesised from engagement (library membership +1.5, playlist count, top-track rank decay, popularity) [`src/music/ingestion.py`]. Affinity is additionally scored by a PU blend (cluster-density + kNN-density + an XGBoost PU classifier trained library-vs-background-pool with hard-negative mining) [`src/music/affinity.py`, `profile_builder.py`]. A stacking regressor (XGB+SVR+CatBoost → Ridge) serves the Music Oracle [`src/music/model_training.py`]. The 3,688 music pseudo-labels feed only the secondary "unified_full" line, never the frozen-fold taste metrics.

The aleatoric-ceiling argument (unquantified). Movies aside, achieved R^2 is low (Figure 5). This is *consistent with*, but not proof of, a per-domain aleatoric ceiling — single-user ratings carry irreducible noise (mood, timing, memory) that no feature can explain. The report is careful not to lean on this as an explanation: the ceiling is **[NOT FOUND IN ARTIFACTS — not quantified in any artifact; the R^2-vs-ceiling figure shows achieved R^2 only, with no ceiling line]**, so low R^2 is reported as a fact and the ceiling only as a plausible, untested account of it.

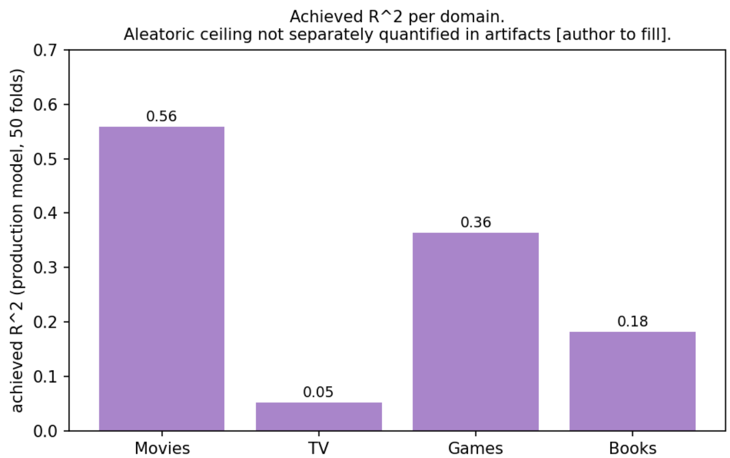

Figure 5: Achieved R^2 per domain (production models). The aleatoric ceiling is not separately estimated in the artifacts.

are correct for their context.

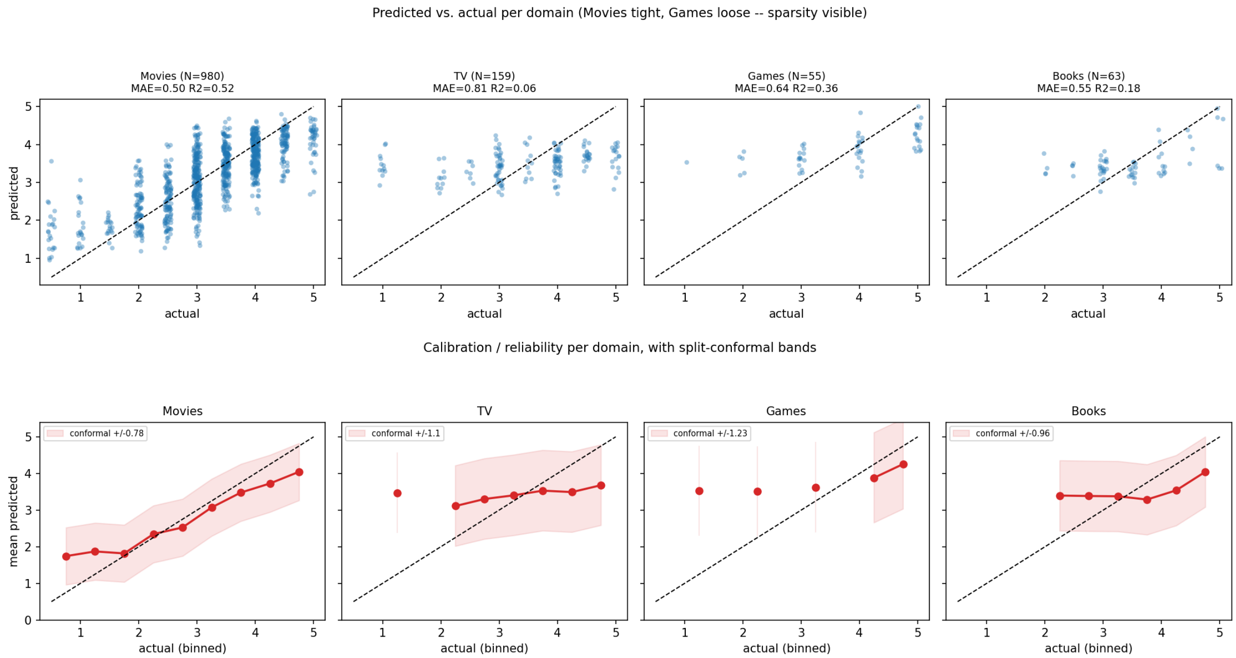

Figure 6: Top: predicted-vs-actual per domain (Movies tight, Games loose — sparsity made visible). Bottom: calibration / reliability with split-conformal bands (± 0.78 movie, ± 1.10 TV, ± 1.23 game, ± 0.96 book) [`standalone_oof.csv`; `active_learning_ranker.py`].

5 The Custom Edge-Penalty Loss

Formulation. To counter regression-to-the-mean at the rating extremes, the project ships a custom XGBoost objective [`src/movies/custom_objectives.py`]:

$$\mathcal{L}(e, r) = \tfrac{1}{2}\, e^2 \cdot \underbrace{\exp\!\big(\alpha_{hi} \max(0, r - 4.0) + \alpha_{lo} \max(0, 1.5 - r)\big)}_{W(r)}, \quad e = \hat{y} - y,\ r = y. \tag{1}$$

Because W depends only on the true rating r, the XGBoost gradient and Hessian are clean: grad $= W(r)\,(\hat{y} - y)$, hess $= W(r)$ [`custom_objectives.py`:27–31]. W equals 1 in the rating interior $[1.5, 4.0]$ and grows toward the "Must Watch" ($r > 4$, rate α_{hi}) and "Hard Pass" ($r < 1.5$, rate α_{lo}) extremes. The tuned values (Optuna, movies) are $\alpha_{hi} = 0.0597$, $\alpha_{lo} = 0.0788$ [`PIPELINE_RUN_LOG.md`].[3]

Ablation (run for this report). Until now the loss was never head-to-head tested. The new ablation [`src/experiments/edge_loss_ablation.py`] holds the movie feature matrix, the 50 frozen folds, and all XGBoost hyperparameters fixed, varying only the objective: **Arm A** `reg:squarederror` vs **Arm B** the edge-penalty loss at the tuned α. No external sample weights are used, so the penalty is the sole reweighting mechanism. Per-item OOF predictions are paired and compared with a Wilcoxon test on the 179 edge items ($r \leq 1.5$ or $r \geq 4.5$) and on all 980 items.

[3] Artifact discrepancy: `latest_metrics.json` records `alpha_hi=null, alpha_lo=null` with the note that the alphas are tuned by the advanced trainer but *not* applied in the frozen-fold Mean Ensemble; `PIPELINE_RUN_LOG.md` says they were persisted to `models/movie/best_params.json`, but that file does not currently exist on disk. The ablation below therefore uses the documented logged values directly.

Table 3: Edge-penalty loss head-to-head, movies, 50 folds [`edge_loss_ablation_results.json`]. **[PILOT]** — directionally correct but *not* statistically significant.

Metric	Arm A (sq-err)	Arm B (edge)	Wilcoxon p
Overall MAE	0.4867	0.4847	0.214
Edge MAE ($\leq$1.5 or $\geq$4.5)	0.8032	0.7935	0.279
Prediction spread (SD)	0.7193	0.7244	–
$\pm 0.5\star$ accuracy	80.0%	79.9%	–
Exact-match at extremes	10.1%	9.5%	–

The verdict is honest and negative-leaning: at the tuned operating point the loss moves overall MAE by -0.0020 ($p = 0.214$) and edge-MAE by -0.0097 ($p = 0.279$), neither significant; prediction spread rises only from 0.719 to 0.724. The reason is visible in the α sweep (Table 4, Figure 7): the mechanism *does* work — as α grows from 0 to 1.0, edge-MAE falls monotonically ($0.803 \to 0.773$) and prediction spread widens ($0.719 \to 0.748$) — but overall MAE bottoms out near the tuned value and then *rises* ($0.487 \to 0.490$). Optuna, which minimises overall error, therefore correctly drove α small, landing in a regime where the loss is nearly indistinguishable from squared error (Figure 8).

Table 4: Edge-penalty α sweep (*symmetric*, $\alpha_{hi} = \alpha_{lo}$), movies, 50 folds [`edge_loss_ablation_results.json`]. **Reconciliation with Table 3:** this sweep is symmetric, so its $\alpha = 0.0597$ row (overall MAE 0.4831) uses $\alpha_{lo} = 0.0597$, whereas Arm B in Table 3 uses the *asymmetric* tuned pair ($\alpha_{hi} = 0.0597, \alpha_{lo} = 0.0788$) and scores 0.4847. Same model, same folds, same averaging — the 0.0016 gap is precisely the larger low-end penalty α_{lo}, not a rounding or run-to-run discrepancy.

α	Overall MAE	Edge MAE	Pred spread
0.0000 (sq-err)	0.4867	0.8032	0.7193
0.0597 (tuned)	0.4831	0.7918	0.7233
0.3000	0.4846	0.7832	0.7303
0.6000	0.4860	0.7750	0.7382
1.0000	0.4902	0.7726	0.7478

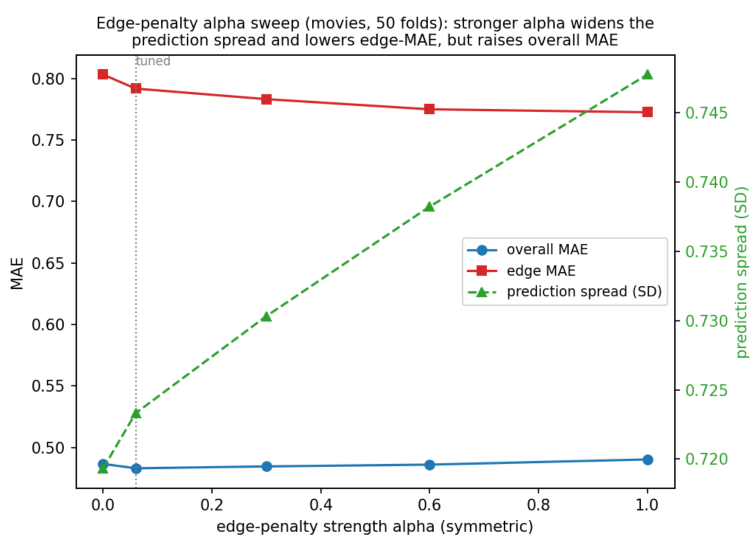

Figure 7: The loss trades overall accuracy for edge accuracy and spread. The tuned α (dotted) sits at the overall-MAE minimum, where the edge benefit is small.

Figure 8: Predicted vs. actual: standard loss (A), edge-penalty at the tuned mild α (B, nearly identical to A), and a strong $\alpha = 0.8$ (visibly less squashed at the extremes). The mechanism un-squashes the extremes only when α is large.

Recommendation. Consistent with the project's kill-what-doesn't-work discipline, the edge-penalty loss is not justified by overall accuracy and only marginally by edge accuracy at its tuned setting. It is defensible as a small, directionally-correct regulariser of the prediction range, but a reviewer asking "how much did the custom loss help?" should be told: *measurably little, and not significantly*, with the sweep offered as evidence the design is sound but the operating point conservative.

6 Unified Cross-Domain Model and Ablations

Architecture ablation. The unified model was grown and then pruned. Table 5 is the rated-only, frozen-fold ablation [`latest_metrics.json`; `UNIFIED_ABLATION_REPORT.md`]. Adding a mean ensemble (XGB+CatBoost) over the aligned base XGBoost improves MAE from 0.5505 to 0.5439 — statistically significant (ΔMAE +0.0066, Cohen's $d = 0.084$, $p = 7.4 \times 10^{-9}$) but *trivial* in effect size [`PIPELINE_RUN_LOG.md`]. The two more complex variants were **[NULL/DROPPED]**: Ridge stacking (MAE 0.5609) and residual heads (MAE 0.6088) both *worsened* performance at this N and were removed (they survive in the table only as legacy 5-fold numbers, explicitly flagged as not frozen-fold comparable). The lesson — complexity did not help — is itself the result.

Table 5: Unified architecture ablation, rated-only, 50 frozen folds [`latest_metrics.json`].

Protocol	MAE	R^2	Effect size	p
Base XGB (Aligned)	0.5505	0.4225	ref	ref
Mean Ensemble (XGB+Cat)	0.5439	0.4384	d=0.084 (trivial)	0.0000
Ridge Stack (removed — legacy 5-fold protocol)	0.5609	0.3837	n/a (legacy)	n/a (legacy)
+Residual Heads (removed — legacy 5-fold protocol)	0.6088	0.2688	n/a (legacy)	n/a (legacy)

Pooling helps the weak, hurts the strong. Figure 9 contrasts each domain's standalone model with its slice of the pooled unified model [`latest_metrics.json` benchmarks vs slices]. Pooling helps the data-poor (book slice MAE 0.540 vs standalone 0.548; TV slice 0.708 vs 0.827) and hurts the data-rich and the sparsest (game slice MAE balloons to 1.048 vs standalone 0.636). This asymmetry is exactly why the deployed system keeps *local* models per domain and uses the unified model for transfer study, not for production scoring of strong domains.

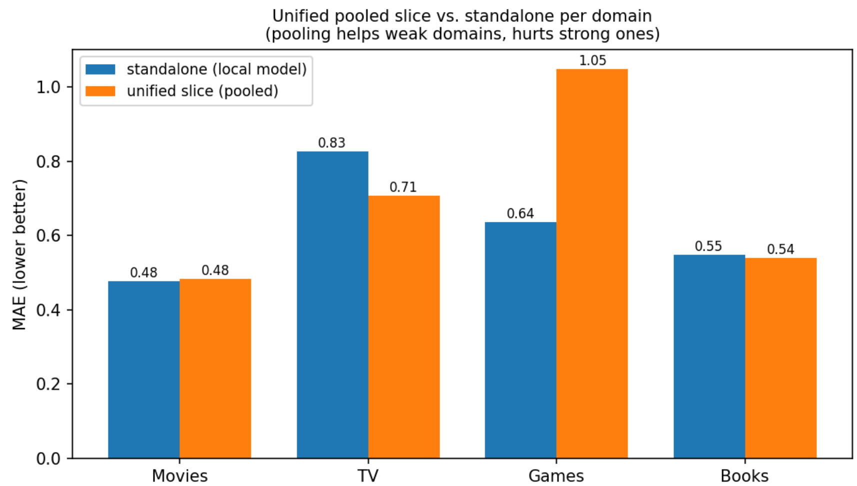

Figure 9: Unified pooled slice vs. standalone per domain (MAE, lower better).

Temporal decay — inert in practice ([NULL/DROPPED]). The Mean Ensemble down-weights ratings by $w_i = \max(e^{-\lambda \Delta t_i}, 0.2)$ with a hardcoded $\lambda = 0.000429$ [`comprehensive_evaluator.py`:20; `unified_utils.py`:77]. *However*, the rating dates this reads are all the placeholder `2000-01-01` (Section 4; verified: 1 unique date across all 4,952 rows), so $\Delta t_i = 0$ for every item and the weights collapse to a uniform 1.0 — the decay does nothing. Consequently the often-quoted "taste half-life ≈ 1,615 days" is merely $\ln 2/\lambda$ for the constant λ and carries *no empirical content*; it is corrected here from a claimed finding to an artifact. Two clarifications keep this honest: (i) the reported Mean Ensemble metrics are unaffected — a uniform weight is just ordinary unweighted training, so Table 5 stands; (ii) the machinery is correct and would yield a real half-life once genuine rating dates are populated (an Optuna path that tunes λ over dates exists in `advanced_unified_model_trainer.py`). The decay is therefore a wired-but-dormant feature, not a result.

Distillation prior — dropped. Feeding the unified model's OOF prediction as an extra feature into the local Games/Books SVRs was tested and [NULL/DROPPED]: Games $p = 0.206$, Books $p = 0.178$ [`distillation_ablation_results.json`]. Table 6. (Note the Games skill nominally rises $0.225 \to 0.234$ with the prior, but the change is not significant, so the discipline drops it; Books skill falls.)

Table 6: Distillation-prior ablation [`distillation_ablation_results.json`]. Both [NULL/DROPPED].

Domain	MAE (no prior)	Skill (no prior)	MAE (with)	p	Verdict
Games	0.6582	0.2253	0.6509	0.2059	DROP
Books	0.5476	0.1078	0.5587	0.1779	DROP

7 The Transfer Study

This is the centrepiece. The question — does taste transfer across domains? — is answered by a grid over (target domain) × (source subset) × (zero-shot / augmented) × (target-data fraction $\in \{0, 0.25, 0.5, 1.0\}$), all on the 50 frozen folds with 300-tree XGBoost in the domain-blind shared space [`src/experiments/transfer_study.py`]. The metric is skill vs the mean baseline; the pre-registered rule for a positive finding is augmented lift > 0 with paired $p < 0.05$ at ≥ 2 fractions.

Headline: transfer into Games is positive. Every positive finding lands in the Games target (Table 7, Figure 11). The strongest is `movie+book→game`: lift@100% = +0.121 with 3/4 fractions significant (zero-shot +0.175, $p = 0.0004$; 50% +0.092, $p = 0.035$; 100% +0.121, $p = 0.0015$) [`transfer_grid_summary.json`]. The augmented learning curve (Figure 12, Games panel) is the money shot: with the source added, skill jumps from the target-alone 0.004 to 0.17 at zero target data and stays above the target-alone line throughout.

Table 7: Positive transfer findings (all into Games), domain-blind, 50 folds [`transfer_grid_summary.json`]. **[CONFIRMED]** after controls (movie+book).

Target	Source	lift@100%	sig. fractions
Games	movie	+0.110	2/4
Games	movie+book	+0.121	3/4
Games	movie+tv	+0.092	3/4
Games	movie+tv+book	+0.094	3/4
Games	tv	+0.102	3/4

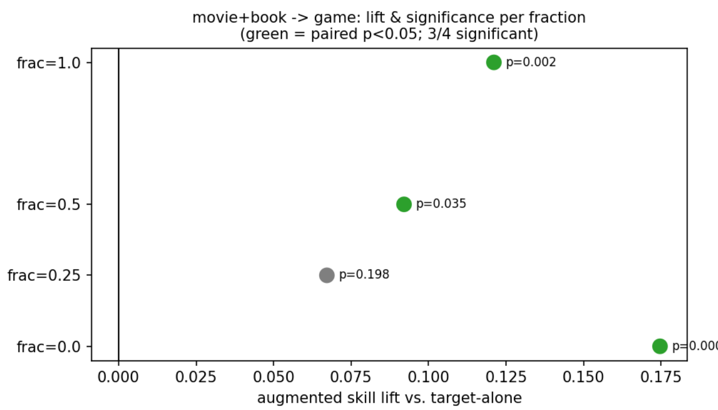

Figure 10: Per-fraction lift and significance for `movie+book`→`game` (green = paired $p < 0.05$): significant at 3/4 fractions [`transfer_grid_summary.json`].

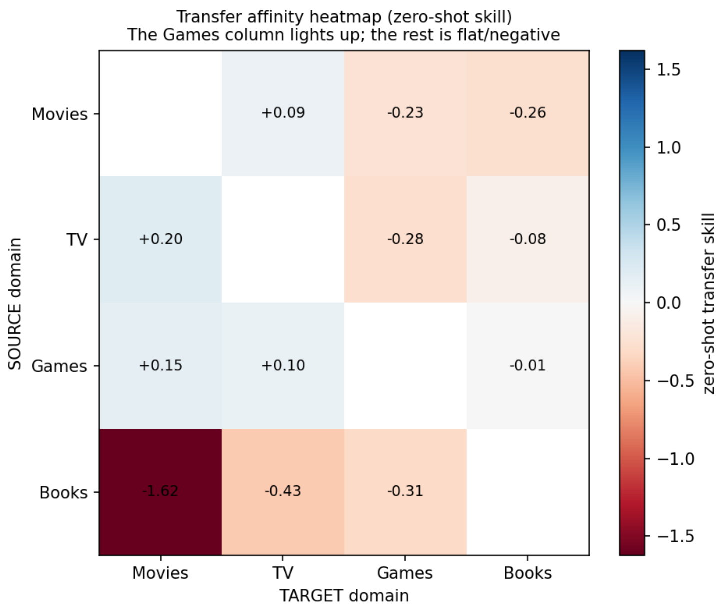

Figure 11: Zero-shot transfer affinity, source (row) $\to$ target (col). The Games column is positive; movie$\to$book is strongly negative (-1.62).

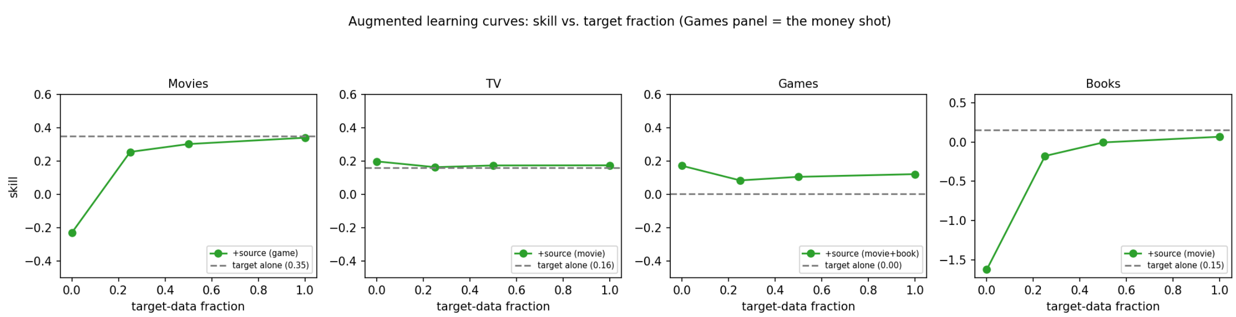

Figure 12: Augmented learning curves. Only Games benefits from a source at low target fractions; Movies and Books are already at or above the augmented line by full data.

Three adversarial controls — the claim narrows but survives.

1. *Domain-identity leak fix.* Diagnostics found `has_*_feats` indicators in the grid and bridge features, letting a model cheat by domain identity (Check 4 **FAIL**) [`PIPELINE_RUN_LOG.md`]. Removing them (domain-blind) drops a spurious "significant" bridge result (ΔMAE +0.032, p = 0.004) to non-significant — and the Games transfer finding *survives*, confirming it is content, not leak.

2. *Prior vs. transfer.* Is the lift just a better constant than the Games mean? No: `movie+book`$\to$`game` beats every constant baseline (MAE 1.081 vs game-/source-/global-mean 1.24/1.23/1.23) and ranks games correctly (Spearman +0.368) $\Rightarrow$ content transfer [`PIPELINE_RUN_LOG.md`; `transfer_prior_control.py`].

3. *Text-length normalization.* Game descriptions are $\sim 5\times$ longer than movie plots (median 1074 vs 214 chars, Check 2), risking a length artifact in the PCA vibe space. Re-embedding all domains to ~ 232 median chars and refitting PCA, `movie+book`$\to$`game` still lifts +0.121 (2/4 significant, zero-shot ρ = +0.36) — **survives** — while the broader source set weakens (tv 0/4, movie 1/4). The robust claim narrows from five source configs to one: `movie+book` [`PIPELINE_RUN_LOG.md`; `text_norm_transfer_control.py`].

This narrowing is summarised in Figure 13. A prior 6-fold pilot had returned **[NULL/DROPPED]** ("no feature-level transfer"); the full 50-fold grid reverses it with the statistical power of 50 vs 6 folds, and the reversal is the *controlled* one above, not an artifact.

Multiple comparisons — the controls, not the grid threshold, carry the claim. The grid tests roughly 30 source configurations $\times$ 4 targets at the pre-registered raw threshold (lift > 0, $p < 0.05$ at ≥ 2 fractions), and *no* family-wise or FDR correction is applied to the grid as a whole — so the broad five-config Games result is treated as *suggestive only*, exactly the "isn't one hit at $p < 0.05$ just luck?" objection. The narrow claim survives that objection for two reasons independent of the grid threshold: (i) `movie+book`$\rightarrow$`game` reaches zero-shot $p = 0.0004$, which clears a Bonferroni correction over ~ 120 cells ($0.05/120 \approx 4 \times 10^{-4}$); and (ii) it passes the two orthogonal controls (prior-vs-transfer and text-normalization), which a multiple-comparisons fluke would not. The honest position: the raw grid p-values are not corrected, the broad set is not promoted, and the single robust finding rests on its low zero-shot p plus the controls.

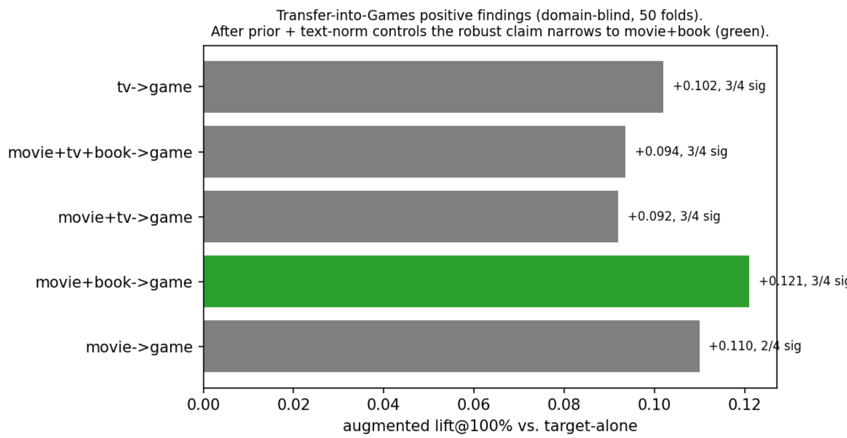

Figure 13: Five broad positive configs collapse to one robust survivor (`movie+book`, green) after the prior and text-normalization controls.

Geometry does not predict transfer (a clean null). Across the 12 ordered domain pairs, centroid distance in the shared space is essentially *anti*-correlated with observed zero-shot skill: Spearman $\rho = -0.18$ [`transfer_grid_summary.json`] (Figure 14). Whatever makes movies+books help games, it is not feature-space proximity — a caution against selecting transfer sources by embedding distance.

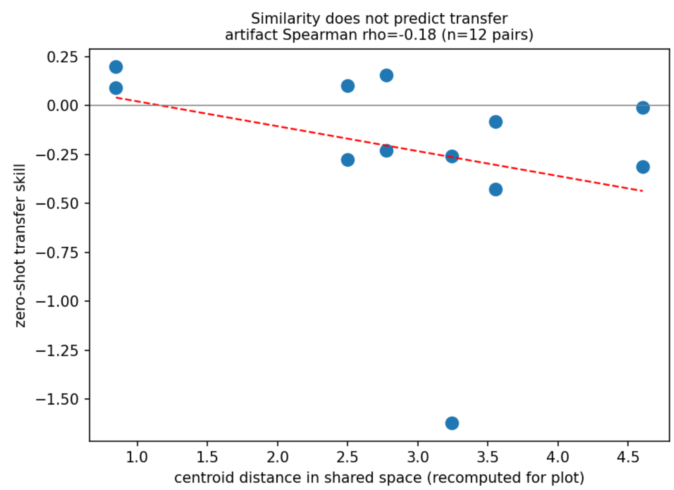

Figure 14: Centroid distance vs. zero-shot skill. The authoritative statistic is the summary's Spearman $\rho = -0.18$ [`transfer_grid_summary.json`], computed by the pipeline over its own centroid/MMD distance. The points here are re-derived for visualisation from standardized PCA+critic+year centroids (Euclidean), a *different* distance metric, so their exact positions need not reproduce -0.18; they are shown only to illustrate the weak, non-positive association. The fitted line is over the plotted points, not the source of the quoted ρ.

8 Entity Bridges and Informative Censoring

Where feature pooling is weak, *item-level* links might carry taste signal. Wikidata linking resolves the 1,264 rated items to QIDs and finds **2,539 links**: 2,085 same-franchise, 441 shared-creator, 13 adaptation [`data/processed/entity_links.csv`; `PIPELINE_RUN_LOG.md`] (Figure 15). Adding bridge features (linked-partner mean rating, franchise mean, has-link) to an XGBoost regressor over the 617 linked-and-rated items improves MAE from 0.5227 to 0.5057, but the effect is **[PILOT]**: ΔMAE $+0.017$, 95% CI $[-0.004, +0.040]$, Wilcoxon $p = 0.182$ [`entity_bridge_results.json`]. The adaptation-only slice is **[NULL/DROPPED]**/degenerate: $N = 22$, ΔMAE 0.000, $p =$ NaN (the OOF bridge feature is constant at this N) [`entity_bridge_results_adaptation.json`].

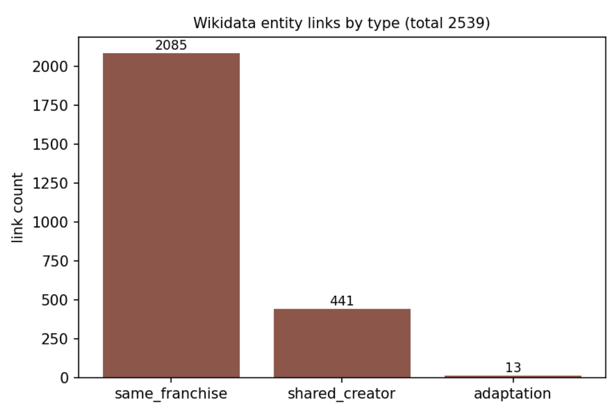

Figure 15: Wikidata links by type (2,539 total): franchise links dominate; adaptations are rare (13) [`entity_links.csv`].

Informative censoring (open direction). A genuinely novel idea sits here: *declining* to consume the adaptation of something you disliked is itself a signal (the "Twilight" case — a disliked book whose film you never rate). The missingness is not at random; it is informative. The data to test it does not yet exist: only **2** of the 13 adaptation links are movie→book, so the censoring-asymmetry figure could not be produced [**[NOT FOUND IN ARTIFACTS — only 2 movie→book adaptation links; censoring asymmetry underpowered]**]. This is framed as an open research direction, not a result.

9 Uncertainty and Active Learning

Each domain gets a split-conformal interval [9] sized to 80% coverage from OOF residual quantiles: ±0.78 (movie), ±1.10 (TV), ±1.23 (game), ±0.96 (book) [`active_learning_ranker.py`]. The interval is *constant* per domain — a by-design property of vanilla split conformal, not a bug; it guarantees marginal, not conditional, coverage. Figure 16 shows empirical coverage against the 80% target, but this is an *in-sample OOF approximation*, which is optimistically biased and should not be read as clean validation: movies (79.8%) and books (85.7%) land near target, but TV *under*-covers at 76.7% and games *over*-covers at 90.9%. The TV under-coverage is a yellow flag, not a pass. True coverage requires a held-out calibration set the project does not have at this N; the constant-width intervals should be treated as rough, not certified.

The active-learning queue ranks unrated items by $Z(\text{uncertainty}) + Z(\text{novelty})$, where uncertainty is ensemble disagreement and novelty is kNN distance in PCA space [`active_learning_ranker.py`]. The current queue is entirely Games (Figure 16) — the sparsest, highest-uncertainty domain — which is the correct thing for the system to want labelled next.

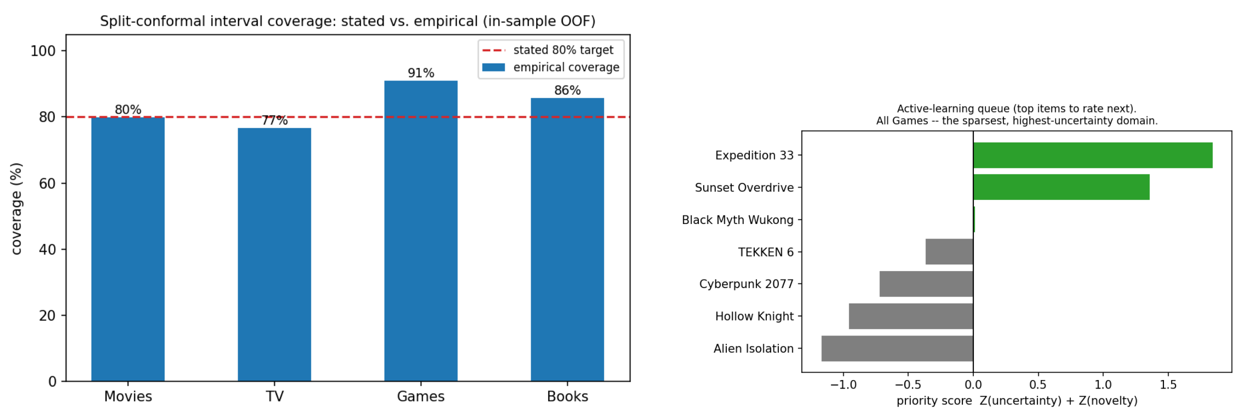

Figure 16: Left: stated vs. empirical conformal coverage per domain. Right: active-learning priority queue (uncertainty/novelty split is not persisted to the queue artifact, so the composite score is shown).

10 Evaluation Methodology and Reproducibility

Frozen-fold registry. A single `RepeatedKFold(10,5,seed=42)` = 50 folds is built once over the rated unified set and stored as a `global_id` → fold-list map [`models/unified/fold_registry.json`; `create_frozen_folds.py`]. Every experiment — benchmarks, ablation, distillation, transfer, the new edge-loss ablation — reuses it, so comparisons are paired at the item level and reproducible by construction.

Single source of truth. `comprehensive_evaluator.py` computes all benchmark/slice/ablation/distillation numbers and writes `latest_metrics.json`; `render_docs.py` renders that JSON into the README and report between markers. No metric is hand-typed into prose. This report follows the same rule — its tables are generated from the artifacts by `paper/make_figures.py`.

Failures caught and corrected (a contribution).

- **Benchmark-inversion / 5× N inflation.** An earlier writer published the unified per-domain *slice* as if it were the standalone benchmark, counting OOF rows (∼ 5× items) as N. Fixed by regenerating genuine standalone OOF deduplicated to one row per item, guarded by assertions that benchmarks $\neq$ slices and $N =$ unique items [`standalone_benchmarks.py`].

- **Leak-faked significance.** The domain-identity leak (Check 4) had produced a "significant" bridge result ($p = 0.004$) that vanished once `has_*_feats` were removed [`PIPELINE_RUN_LOG.md`].
- **Stale-artifact drift.** The shipped music model expected 167 features while the regenerated feature matrix had 127, crashing the Music Oracle; the model was retrained to the current 127-feature space and a feature-count guard was added [`src/music/model_training.py`]. A concrete instance of the drift the single-source-of-truth discipline exists to prevent.
- **Superseded pilots.** The 6-fold transfer pilot (**[NULL/DROPPED]**) and the capped 1,173-link bridge run are retained in the log but explicitly marked superseded; only the latest 50-fold / 2,539-link numbers are used.

11 Limitations

- **Aleatoric ceiling, unquantified.** Low R^2 in TV/Books is partly irreducible single-user noise, but the ceiling is not estimated (**[NOT FOUND IN ARTIFACTS — ceiling value]**).
- **Single user.** $N = 1$; nothing here claims to generalise across people.
- **Small-N domains.** Games/Books/TV have $N < 200$; significance is fragile and the transfer reversal hinged on getting full statistical power.
- **PU pseudo-labels.** Music ratings are synthesised from engagement, not stated; the music line is kept secondary for this reason.
- **Production text not normalized.** The text-length fix exists only as a control; production embeddings still see un-normalized text.
- **Underpowered bridges.** The entity-bridge effect is a non-significant nudge ($p = 0.182$) and the adaptation slice is degenerate ($N = 22$).

12 Conclusion and Future Work

PMIH establishes, on honest frozen-fold evaluation, that single-user taste *can* transfer into a data-poor domain (Games) from richer ones (Movies+Books), that this survives three adversarial controls, and that feature-space geometry does *not* predict which transfers work. It establishes equally clearly what does *not* help: Ridge stacking and residual heads (over-fit at small N), the distillation prior, and — newly — the custom edge-penalty loss at its tuned operating point.

Future work. (1) **YouTube as a sixth, implicit-feedback domain**: watch-history is ingested and enriched [`src/youtube/`] but deliberately *not* modelled — watch-time is a weak, confounded label and including it now would import noise into the clean five-domain transfer story; it is scoped for a separate implicit-feedback study. (2) **Full-grid confirmations** for the borderline source configs that the controls weakened. (3) **The censoring experiment**, once enough movie$\leftrightarrow$book adaptation pairs are rated to make it powered. (4) **Production text normalization**, promoting the control fix into the live embedding pipeline.

A Appendix A: Figures and Tables Generated

All regenerated by `python -m paper.make_figures` and `python -m src.experiments.edge_loss_ablation`. Files in `paper/figures/` and `paper/tables/`.

Artifact	Source data
`rating_distributions.png`	`training_features.csv`
`feature_coverage_heatmap.png`	`training_features.csv`
`pred_vs_actual_per_domain.png`	`standalone_oof.csv`
`calibration_per_domain.png`	`standalone_oof.csv`, conformal widths
`r2_achieved_per_domain.png`	`latest_metrics.json` (ceiling omitted)
`slice_vs_standalone.png`	`latest_metrics.json`
`transfer_affinity_heatmap.png`	`transfer_grid_summary.json`
`transfer_learning_curves.png`	`transfer_grid_summary.json`
`games_forest_plot.png`	`transfer_grid_summary.json`
`similarity_vs_transfer.png`	`transfer_grid_summary.json` + recomputed centroids
`transfer_controls_narrowing.png`	`transfer_grid_summary.json`
`umap_domain_and_rating.png`	`training_features.csv` (UMAP)
`coral_separability.png`	`PIPELINE_RUN_LOG.md` Check 3
`text_length_per_domain.png`	`PIPELINE_RUN_LOG.md` Check 2 (medians)
`entity_link_counts.png`	`entity_links.csv`
`conformal_coverage.png`	`standalone_oof.csv`
`active_learning_queue.png`	`ACTIVE_LEARNING_QUEUE.md`
`edge_loss_pred_vs_actual_3panel.png`	`edge_loss_ablation_results.json`
`edge_loss_calibration.png`	edge-loss ablation
`edge_loss_alpha_sweep.png`	edge-loss ablation
censoring_asymmetry	*NOT generated — only 2 movie→book adaptations*
`per_domain_summary.tex`	`training_features.csv`
`benchmark.tex`	`latest_metrics.json`, `production_oof.csv`
`edge_loss.tex`, `alpha_sweep.tex`	`edge_loss_ablation_results.json`
`architecture_ablation.tex`	`latest_metrics.json`
`distillation.tex`	`distillation_ablation_results.json`
`transfer_positive.tex`	`transfer_grid_summary.json`

B Appendix B: Claims Audit

Every quantitative claim, its source, and its tier.

Claim	Source	Tier
Movies MAE 0.478, $R^2$0.558, Acc 79.1%	`latest_metrics.json`	**[CONFIRMED]**
TV MAE 0.827, $R^2$0.052	`latest_metrics.json`	**[CONFIRMED]**
Games MAE 0.636, $R^2$0.364 ($N = 55$)	`latest_metrics.json`	**[CONFIRMED]**
Books MAE 0.548, $R^2$0.182 ($N = 63$)	`latest_metrics.json`	**[CONFIRMED]**
Skill: movie 0.362/tv 0.075/game 0.207/book 0.126	`production_oof.csv,standalone_oof.csv`	**[CONFIRMED]**
Mean ensemble ΔMAE +0.0066, d=0.084, p=7.4e−9	`latest_metrics.json,PIPELINE_RUN_LOG.md`	**[CONFIRMED]**(trivial)
Ridge stack / residual heads worse	`latest_metrics.json`	**[NULL/DROPPED]**
Temporal λ = 0.000429 hardcoded; "half-life" inert (placeholder dates)	`comprehensive_evaluator.py`, `training_features.csv`	**[NULL/DROPPED]**
Distillation prior: game p=0.206, book p=0.178	`distillation_ablation_results.json`	**[NULL/DROPPED]**
movie+book→game lift@100 +0.121, 3/4 sig	`transfer_grid_summary.json`	**[CONFIRMED]**
All positive transfers land in Games	`transfer_grid_summary.json`	**[CONFIRMED]**
Prior control: Spearman +0.368, beats constants	`PIPELINE_RUN_LOG.md`	**[CONFIRMED]**
Text-norm: lift +0.121 survives, narrows to movie+book	`PIPELINE_RUN_LOG.md`	**[CONFIRMED]**
Similarity↔transfer $\rho = -0.18$ (n=12)	`transfer_grid_summary.json`	**[CONFIRMED]**(null)
Entity links 2539 (2085/441/13)	`entity_links.csv`	**[CONFIRMED]**

Bridge ΔMAE +0.017, $p = 0.182$ (N=617)	`entity_bridge_results.json`	[PILOT]
Adaptation bridge N=22, degenerate	`entity_bridge_results_adaptation.json`	[NULL/DROPPED]
Edge loss: Δoverall MAE -0.0020 (p=0.21)	`edge_loss_ablation_results.json`	[PILOT](n.s.)
Edge loss: Δedge MAE -0.0097 (p=0.28)	`edge_loss_ablation_results.json`	[PILOT](n.s.)
Edge loss sweep: edge-MAE ↓, spread ↑, overall MAE ↑	`edge_loss_ablation_results.json`	[CONFIRMED](mechanism)
Tuned α_{hi}=0.0597, α_{lo}=0.0788	`PIPELINE_RUN_LOG.md`	[CONFIRMED](see footnote
Conformal widths 0.78/1.10/1.23/0.96	`active_learning_ranker.py`	[CONFIRMED]
unified_full MAE 0.411 (N=4952, +music)	`latest_metrics.json`	[PILOT](secondary, not frozen

C Appendix C: Open Gaps (`[NOT FOUND IN ARTIFACTS]`)

1. **Aleatoric ceiling per domain** — no quantified ceiling in any artifact; the R^2-vs-ceiling figure shows achieved R^2 only.

2. **Tuned-α persistence** — `models/movie/best_params.json` is referenced by `PIPELINE_RUN_LOG.md` but absent on disk; `latest_metrics.json` stores the alphas as `null`. The logged values (0.0597/0.0788) were used for the ablation.

3. **Censoring asymmetry** — only 2 movie→book adaptation links exist; the figure cannot be produced until more such pairs are rated.

4. **Active-learning uncertainty/novelty split** — the queue artifact stores only the composite priority, not the two components; the scatter of uncertainty × novelty could not be drawn from the saved artifact.

5. **Per-domain feature-coverage exact percentages** for raw critic sub-scores are approximated by non-zero presence, not by source-level null tracking.

6. **Real rating dates** — `rating_date` is NaN-filled to the placeholder `2000-01-01` for all 4,952 rows [`unified_feature_engineering.py`:272], so temporal decay is inert and the "taste half-life" has no empirical content (Section 6). Populating genuine rating timestamps would activate the (already wired) decay and yield a real half-life.

7. **Held-out conformal calibration set** — conformal coverage is only an in-sample OOF approximation; a proper calibration split (absent at this N) is needed to certify coverage, which currently mis-targets for TV (76.7%) and games (90.9%).

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